

Custom Loss Function Explanation

1 Combined Loss Function

The total loss function is defined as:

$$L = \beta L_{MCC} + \alpha L_{Focal} + \gamma L_{Dice} \quad (1)$$

where:

- L_{MCC} = Matthews Correlation Coefficient Loss
- L_{Focal} = Focal Loss
- L_{Dice} = Dice Loss
- α, β, γ = weighting coefficients

This loss combines classification balance, hard sample learning, and segmentation quality.

2 MCC Loss

The Matthews Correlation Coefficient (MCC) is:

$$MCC = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (2)$$

where:

- TP = True Positive
- TN = True Negative
- FP = False Positive
- FN = False Negative

The MCC loss is defined as:

$$L_{MCC} = 1 - MCC \quad (3)$$

Value Range

$$-1 \leq MCC \leq 1$$

So,

$$0 \leq L_{MCC} \leq 2$$

Lower values indicate better performance.

3 Focal Loss

Focal Loss is defined as:

$$L_{Focal} = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (4)$$

where:

- α = class balancing factor
- γ = focusing parameter
- p_t = predicted probability of the true class

Parameter Values

Typical values:

- $\alpha = 0.25$ to 0.75
- $\gamma = 1$ to 5

Common setting:

$$\alpha = 0.25, \quad \gamma = 2$$

Value Range

$$0 \leq L_{Focal} < \infty$$

Lower is better.

4 Dice Loss

Dice coefficient is:

$$Dice = \frac{2|X \cap Y|}{|X| + |Y|} \quad (5)$$

Dice Loss is:

$$L_{Dice} = 1 - Dice \quad (6)$$

Value Range

$$0 \leq Dice \leq 1$$

Thus,

$$0 \leq L_{Dice} \leq 1$$

Lower values indicate better overlap.

5 Weighting Parameters

The coefficients α , β , and γ control the importance of each loss term.

Recommended initial values:

$$\alpha = 1, \quad \beta = 1, \quad \gamma = 1 \tag{7}$$

Possible tuning:

- Increase α for severe class imbalance
- Increase β for balanced classification emphasis
- Increase γ for segmentation emphasis

6 Optimization Objective

The optimization objective is:

$$\min_{\alpha, \beta, \gamma} L \tag{8}$$

where the goal is to minimize the total loss during training.

7 Interpretation of Good Values

A good model should have:

- $L_{MCC} \rightarrow 0$
- $L_{Focal} \rightarrow 0$
- $L_{Dice} \rightarrow 0$
- Total Loss $\rightarrow 0$

Lower total loss indicates better model learning.